252PS032

Department of Computer Science and Engineering
National Institute of Technology Karnataka, Surathkal

Mid-Semester Examination (February 2026)

Course Name: Quantum Cryptography
Semester: II
Duration: 1 Hour 30 Minutes
Note: Answer all the questions.

Course Code: CS826
Course: M.Tech. (CSE-IS)
Marks: 25

Q1. (a) A qubit is in the following state: [2]

$$\begin{pmatrix} \sqrt{3}/2 \\ 1/2 \end{pmatrix}$$

If you measure this qubit in the Z-basis $\{|0\rangle, |1\rangle\}$, what states can you get and with what probabilities?

(b) Consider

$$|a\rangle = \frac{3+i\sqrt{3}}{4}|0\rangle + \frac{1}{2}|1\rangle,$$
$$|b\rangle = \frac{1}{4}|0\rangle + \frac{\sqrt{15}}{4}|1\rangle.$$

(i) Find $\langle a|b\rangle$. [1]
(ii) Find $\langle b|a\rangle$. [1]
(iii) What is the relationship between your answers to parts (a) and (b)? [1]

(c) Consider a qubit in the following state [4]

$$|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle.$$

Consider measuring this qubit in the Y-basis $\{|i\rangle, |-i\rangle\}$ and the orthonormal basis $\{|a\rangle, |b\rangle\}$, where

$$|a\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle,$$
$$|b\rangle = \frac{i}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$$

(i) Calculate $\langle i|\psi\rangle$.
(ii) Calculate $\langle -i|\psi\rangle$.
(iii) If you measure the qubit in the Y-basis, what states can you get and with what probabilities?
(iv) Calculate $\langle a|\psi\rangle$.
(v) Calculate $\langle b|\psi\rangle$.
(vi) If you measure the qubit in the $\{|a\rangle, |b\rangle\}$ basis, what states can you get and with what probabilities?

Q2. (a) The following two states are opposite points on the Bloch sphere:

$$|a\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle,$$
$$|b\rangle = \frac{i}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle.$$

So, we can measure relative to them. Now consider a qubit in the state

$$|\varphi\rangle = \frac{1}{2}|0\rangle - \frac{\sqrt{3}}{2}|1\rangle.$$

(i) Write the qubit's state in terms of $|a\rangle$ and $|b\rangle$. [2]

(ii) If you measure the qubit in the basis $\{|a\rangle, |b\rangle\}$, what states can you get and with what probabilities? [2]

(b) Consider the following two states

$$|a\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle,$$
$$|b\rangle = \frac{i}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle.$$

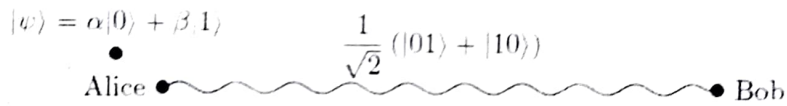
Prove these are opposite points of the Bloch sphere by finding their points in spherical coordinates (θ_a,φ_a) and (θ_b,φ_b). Verify that $\theta_b = \pi - \theta_a$ and $\varphi_b = \varphi_a + \pi$, which means they lie on opposite points of the Bloch sphere. [2]

Q3. Consider the following state of two qubits:

$$\frac{\sqrt{3}}{2\sqrt{2}}|00\rangle + \frac{1}{2\sqrt{2}}|01\rangle + \frac{1}{2\sqrt{2}}|10\rangle + \frac{\sqrt{3}}{2\sqrt{2}}|11\rangle$$

If you measure the left qubit, what outcomes can you get, what are the corresponding probabilities of those outcomes, and what does the state collapse to for each outcome? Is this state a product state, partially entangled state, or maximally entangled state? [1+1+1+1]

Q4 Alice wants to teleport a qubit in an unknown state $|\psi\rangle = \alpha|0\rangle+\beta|1\rangle$ to Bob. Instead of sharing two entangled qubits in the $|\Phi+\rangle$ state, they share two entangled qubits in the $|\Psi+\rangle$ state:



Altogether, the initial state of the system is

$$|\psi\rangle|\Psi^+\rangle = (\alpha|0\rangle + \beta|1\rangle)\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$
$$= \frac{1}{\sqrt{2}}(\alpha|001\rangle + \alpha|010\rangle + \beta|101\rangle + \beta|110\rangle).$$

So, the left two qubits are Alice's, and the right qubit is Bob's.

(i) Show that if Alice applies CNOT to her two qubits, followed by H to her left qubit, the state of the system becomes [2]

$$\frac{1}{2}\big[|00\rangle(\beta|0\rangle + \alpha|1\rangle) + |01\rangle(\alpha|0\rangle + \beta|1\rangle)$$
$$+ |10\rangle(-\beta|0\rangle + \alpha|1\rangle) + |11\rangle(\alpha|0\rangle - \beta|1\rangle)\big].$$

(ii) Next, Alice measures both of her qubits. What values can she get, with what probabilities, and what does the state collapse to in each case? [2]

(ii) Finally, Alice tells Bob the results of her measurement. For each possible result, what should Bob do to his qubit so that it is $\alpha|0\rangle+\beta|1\rangle$, the state that Alice wanted to teleport to him? [2]